

Matrix Theory - Fall 2026

Practice Problem Set 06

1. Consider the matrix

$$A = \begin{bmatrix} 2 & -1 & 0 & 3 \\ 4 & 5 & -2 & 1 \\ 0 & 3 & 6 & -4 \end{bmatrix}.$$

Express A as a collection of its column vectors, and as a collection of its row vectors. Then, for each of the following, determine whether the operation is defined, and if so, compute it:

- (a) A^T
- (b) $A^T A$
- (c) AA^T
- (d) $A^T A^T$
- (e) $A^T A^T A$

2. Let $A = \begin{bmatrix} 1 & 0 & 2 \\ 3 & 1 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} 0 & 1 \\ 2 & 1 \\ 1 & 0 \end{bmatrix}$.

- (a) Check that AB is defined, find its size, and compute it.
- (b) Check that BA is defined, find its size, and compute it.
- (c) Can the columns of A span $\mathbb{R}^2$? Can the columns of B span $\mathbb{R}^3$? Explain.

3. Let $C = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ and $D = \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}$.

- (a) Compute the Hadamard product $C \circ D$.
- (b) Compute the ordinary matrix product CD .
- (c) These should be different. Now suppose instead C were 2×3 and D were 3×2 . Would $C \circ D$ still be defined? Would CD still be defined?

4. Let $A \in \mathbb{R}^{m \times m}$ have nonzero column vectors $\mathbf{a}_1, \dots, \mathbf{a}_m$. Show that the following are equivalent:

- 1. The columns of A are linearly independent.
- 2. The only solution to $A\mathbf{x} = \mathbf{0}$ is $\mathbf{x} = \mathbf{0}$.
- 3. The equation $A\mathbf{x} = \mathbf{b}$ has a unique solution for every $\mathbf{b} \in \mathbb{R}^m$.

(Hint: Use the fact that determining x is equivalent to determining the coefficients of the linear combination.)

5. Let A be a square matrix with columns $\mathbf{a}_1, \dots, \mathbf{a}_n$. Suppose $\|\mathbf{a}_i\| \neq 0 \forall i$. Determine the matrix F such that the columns of FA have unit L_2 norm.

6. Let $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$ be nonzero vectors. What is the sum of diagonal entries of $\mathbf{u}\mathbf{v}^T$?
7. **True or False.** For each statement about $A, B \in \mathbb{R}^{n \times n}$, say whether it is true or false, and justify your answer in one or two sentences.
- (a) The set of all $n \times n$ upper-triangular matrices is a subspace of $\mathbb{F}^{n \times n}$.
 - (b) If A is symmetric, then A^2 (meaning AA) is symmetric.
 - (c) If A and B are both symmetric, then AB is symmetric.
 - (d) Every matrix in $\mathbb{R}^{n \times n}$ can be written uniquely as a sum of a symmetric matrix and a skew-symmetric matrix.
 - (e) The zero matrix is the only matrix that is simultaneously symmetric and skew-symmetric.
8. Overview: The formal, coordinate-free definition of the transpose says that A^T is the unique matrix satisfying $\langle \mathbf{A}\mathbf{x}, \mathbf{y} \rangle = \langle \mathbf{x}, A^T \mathbf{y} \rangle$ for all $\mathbf{x}, \mathbf{y}$. This problem asks you to check that identity by direct computation on a specific matrix.

Problem Statement: Let $A = \begin{bmatrix} 2 & -1 & 0 \\ 1 & 0 & 3 \end{bmatrix}$, $\mathbf{x} = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix} \in \mathbb{R}^3$, and $\mathbf{y} = \begin{bmatrix} 1 \\ -2 \end{bmatrix} \in \mathbb{R}^2$.

- (a) Compute $\mathbf{A}\mathbf{x}$ and hence $\langle \mathbf{A}\mathbf{x}, \mathbf{y} \rangle$.
 - (b) Compute A^T , then $A^T \mathbf{y}$, and hence $\langle \mathbf{x}, A^T \mathbf{y} \rangle$.
 - (c) Confirm that your two answers agree.
9. Let $A = \begin{bmatrix} 3 & 4 \\ 0 & 1 \end{bmatrix}$ and $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$.
- (a) Find the symmetric part $S = \frac{1}{2}(A + A^T)$ and the skew-symmetric part $K = \frac{1}{2}(A - A^T)$ of A .
 - (b) Expand $\mathbf{x}^T \mathbf{A}\mathbf{x}$ as a polynomial in x_1, x_2 .
 - (c) Expand $\mathbf{x}^T S \mathbf{x}$ as a polynomial in x_1, x_2 and confirm it equals your answer to part (b).
 - (d) Expand $\mathbf{x}^T K \mathbf{x}$ directly and confirm it is identically 0, for every x_1, x_2 .
 - (e) By evaluating $\mathbf{x}^T \mathbf{A}\mathbf{x}$ at $\mathbf{x} = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$, show that A (equivalently S) is *not* positive definite.
10. Overview: The *cross product* is an operation you will study formally in vector calculus. In $\mathbb{R}^2$, for $\mathbf{u}, \mathbf{v} \in \mathbb{R}^2$, it is the scalar

$$\mathbf{u} \times \mathbf{v} := u_1 v_2 - u_2 v_1.$$

In $\mathbb{R}^3$, for $\mathbf{w} = [w_1 \ w_2 \ w_3]^T$ and $\mathbf{v} = [v_1 \ v_2 \ v_3]^T$, it instead produces a *vector*:

$$\mathbf{w} \times \mathbf{v} := \begin{bmatrix} w_2 v_3 - w_3 v_2 \\ w_3 v_1 - w_1 v_3 \\ w_1 v_2 - w_2 v_1 \end{bmatrix}.$$

For now, here is an interesting result connecting it to skew-symmetric matrices, which you will encounter again in future courses.

A skew-symmetric matrix $K \in \mathbb{R}^{3 \times 3}$ is completely determined by three numbers. This is not a coincidence: for $\mathbf{w} = [w_1 \ w_2 \ w_3]^T \in \mathbb{R}^3$, define

$$\hat{\mathbf{w}} := \begin{bmatrix} 0 & -w_3 & w_2 \\ w_3 & 0 & -w_1 \\ -w_2 & w_1 & 0 \end{bmatrix}.$$

One can check that $\hat{\mathbf{w}} \mathbf{v} = \mathbf{w} \times \mathbf{v}$ for every $\mathbf{v} \in \mathbb{R}^3$, that is, left multiplication by the skew-symmetric matrix $\hat{\mathbf{w}}$ is the cross product with $\mathbf{w}$. This is exactly why skew-symmetric matrices show up in rigid-body mechanics and robotics: the velocity of a point at position $\mathbf{r}$ on a body spinning with angular velocity $\boldsymbol{\omega}$ is $\mathbf{v} = \boldsymbol{\omega} \times \mathbf{r} = \hat{\boldsymbol{\omega}} \mathbf{r}$.

11. Overview: In a plane electromagnetic wave travelling through vacuum, the electric field $\mathbf{E}$ and magnetic field $\mathbf{B}$ oscillate perpendicular to *each other*, and the wave's energy propagates in a third direction, perpendicular to *both* of them. This direction (and rate) of energy flow is given by the *Poynting vector*

$$\mathbf{S} = \frac{1}{\mu_0} \mathbf{E} \times \mathbf{B},$$

so at any instant, $\{\mathbf{E}, \mathbf{B}, \mathbf{S}\}$ is a mutually orthogonal triad of directions in space. This is exactly the kind of orthogonal set you built with Gram-Schmidt, and the cross product linking them is, once again, multiplication by the skew-symmetric matrix $(\hat{\cdot})$ from the previous problem.

Problem Statement: At some point and instant, a plane wave has (suitably scaled, and ignoring μ_0 for simplicity)

$$\mathbf{E} = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} \text{ V/m}, \quad \mathbf{B} = \begin{bmatrix} 2 \\ -1 \\ 0 \end{bmatrix} \text{ T}.$$

- Verify, using the inner product, that $\mathbf{E} \perp \mathbf{B}$.
 - Using the skew-symmetric matrix $\hat{\mathbf{E}}$ from the previous problem (built from $\mathbf{E}$ instead of $\mathbf{w}$), compute $\hat{\mathbf{E}} \mathbf{B}$. This is (proportional to) the Poynting vector $\mathbf{S}$, the direction in which the wave's energy travels.
 - Verify directly that $\mathbf{S} \perp \mathbf{E}$ and $\mathbf{S} \perp \mathbf{B}$. In words: the energy flows in a direction perpendicular to *both* fields, not along either of them.
 - Normalize $\mathbf{E}, \mathbf{B}, \mathbf{S}$ to obtain unit vectors $\hat{\mathbf{e}}, \hat{\mathbf{b}}, \hat{\mathbf{s}}$.
12. Overview: Just like any square matrix, the field-gradient matrix $\nabla \mathbf{E}$ (where $(\nabla \mathbf{E})_{ij} = \partial E_i / \partial x_j$) splits into a symmetric part $S = \frac{1}{2}(\nabla \mathbf{E} + (\nabla \mathbf{E})^T)$ and a skew-symmetric part $K = \frac{1}{2}(\nabla \mathbf{E} - (\nabla \mathbf{E})^T)$.

The skew-symmetric part K is not just algebra here: it encodes the curl of $\mathbf{E}$. In two dimensions,

$$K_{21} = \frac{1}{2} \left(\frac{\partial E_2}{\partial x_1} - \frac{\partial E_1}{\partial x_2} \right) = \frac{1}{2} (\nabla \times \mathbf{E})_z.$$

This matters because of Faraday's law, $\nabla \times \mathbf{E} = -\partial \mathbf{B}/\partial t$. A purely electrostatic field is curl-free (since $\mathbf{E} = -\nabla \phi$ for a scalar potential ϕ), so it must have $K = \mathbf{0}$. In other words, the skew-symmetric part of $\nabla \mathbf{E}$ tells us whether a field could be electrostatic, or whether it must come with a changing magnetic field.

Problem Statement: A 2-D electric field measured near a point has field-gradient matrix (in V/m per m)

$$\nabla \mathbf{E} = \begin{bmatrix} 2 & 3 \\ 1 & 4 \end{bmatrix} \quad \left(\text{i.e., } \frac{\partial E_1}{\partial x_1} = 2, \frac{\partial E_1}{\partial x_2} = 3, \frac{\partial E_2}{\partial x_1} = 1, \frac{\partial E_2}{\partial x_2} = 4 \right).$$

- (a) Compute the symmetric part S and skew-symmetric part K of $\nabla \mathbf{E}$.
 - (b) Compute $(\nabla \times \mathbf{E})_z = \partial E_2/\partial x_1 - \partial E_1/\partial x_2$ directly from $\nabla \mathbf{E}$, and verify that it equals $2K_{21}$.
 - (c) Based on K , could this $\mathbf{E}$ be a purely electrostatic field (i.e., $\mathbf{E} = -\nabla \phi$ for some scalar potential ϕ)? Using Faraday's law $\nabla \times \mathbf{E} = -\partial \mathbf{B}/\partial t$, what does your answer imply about the magnetic field in this region?
13. Overview: For a linear, passive, *reciprocal* n -port electrical network (built only from resistors, capacitors, inductors, and ideal transformers), the port impedance matrix Z in $\mathbf{V} = Z\mathbf{I}$ always satisfies $Z = Z^T$; this is the reciprocity theorem. A network containing a non-reciprocal element, such as a gyrator or a magnetized-ferrite circulator, can have $Z \neq Z^T$. The symmetric part of Z represents the equivalent reciprocal behaviour of the network, while the skew-symmetric part measures exactly how much, and in what way, reciprocity is being violated.

Problem Statement: A two-port network is measured to have impedance matrix (in Ω)

$$Z = \begin{bmatrix} 50 & 10 \\ 6 & 50 \end{bmatrix}.$$

- (a) Compute the symmetric part Z_{sym} and the skew-symmetric part Z_{skew} of Z .
 - (b) If this network were perfectly reciprocal, what values would Z_{12} and Z_{21} both have to take? Compare with Z_{sym} .
 - (c) Explain, in one sentence, what the nonzero entries of Z_{skew} tell you about this particular network.
14. Overview: Treating the $m \times n$ matrix A as a stack of m row vectors $\mathbf{a}_1^T, \dots, \mathbf{a}_m^T \in \mathbb{R}^n$, the product AA^T turns out to be nothing but a table of *inner products* of these rows with each other. This lets us recover a purely vector-space fact (non-negativity of $\langle \mathbf{u}, \mathbf{u} \rangle$) as a matrix-theoretic one.
- Problem Statement: Let A be any $m \times n$ matrix with rows $\mathbf{a}_1^T, \dots, \mathbf{a}_m^T$.
- (a) Show that $(AA^T)_{ij} = \langle \mathbf{a}_i, \mathbf{a}_j \rangle$ for every i, j , directly from the definition of matrix multiplication.
 - (b) Deduce that the diagonal entries of AA^T are $\|\mathbf{a}_i\|^2 \geq 0$, and hence that AA^T is symmetric.
15. Let B be an $m \times n$ matrix with columns $\mathbf{b}_1, \dots, \mathbf{b}_n$. Show that $\{\mathbf{b}_1, \dots, \mathbf{b}_n\}$ is an orthonormal set if and only if $B^T B = I_n$.
16. **True or False.** Let A be an $m \times n$ matrix with columns $\mathbf{b}_1, \dots, \mathbf{b}_n \in \mathbb{R}^m$. Justify each answer in one or two sentences.

- (a) If $\mathbf{b}_1, \dots, \mathbf{b}_n$ are pairwise orthogonal (not necessarily unit vectors), then $A^T A$ is a diagonal matrix.
- (b) If $A^T A$ is a diagonal matrix, then $\mathbf{b}_1, \dots, \mathbf{b}_n$ must be pairwise orthogonal.
- (c) The diagonal entries of $A^T A$ equal $\|\mathbf{b}_1\|, \dots, \|\mathbf{b}_n\|$ (the norms themselves, not their squares).
- (d) AA^T and $A^T A$ always have the same size.
17. For a square matrix $M \in \mathbb{F}^{n \times n}$, recall the *trace* is $\text{tr}(M) = \sum_{i=1}^n m_{ii}$. Let $A \in \mathbb{F}^{m \times n}$ and $B \in \mathbb{F}^{n \times m}$, so that both AB (size $m \times m$) and BA (size $n \times n$) are defined, even though they may be different sizes. Prove that

$$\text{tr}(AB) = \text{tr}(BA).$$

(Hint: write $\text{tr}(AB)$ as a double sum and swap the order of summation.)

18. For a square matrix $M \in \mathbb{F}^{n \times n}$, how can we find the **sum of the squares of all its elements**? Express your answer in terms of a standard matrix operation.

More explicitly, if $M = [m_{ij}]$, find an expression for

$$\sum_{i=1}^n \sum_{j=1}^n |m_{ij}|^2.$$

(Hint: think of trace of a matrix)

19. Suppose p is a polynomial of degree $n-1$ or less, given by $p(t) = c_1 + c_2 t + \dots + c_n t^{n-1}$. Its derivative (with respect to t), $p'(t)$, is a polynomial of degree $n-2$ or less, given by $p'(t) = d_1 + d_2 t + \dots + d_{n-1} t^{n-2}$. We can represent these polynomials as column vectors of their coefficients: $\mathbf{c} \in \mathbb{R}^n$ and $\mathbf{d} \in \mathbb{R}^{n-1}$. Find a matrix D for which $\mathbf{d} = D\mathbf{c}$. Give the entries of D and specify the dimensions of D .

20. Assuming the matrix

$$K = \begin{bmatrix} I & A^T \\ A & 0 \end{bmatrix}$$

makes sense, which of the following statements must be true? ('Must be true' means that it follows with no additional assumptions.)

- (a) K is square.
- (b) A is square or wide.
- (c) K is symmetric, i.e., $K^T = K$.
- (d) The identity and zero submatrices in K have the same dimensions.
- (e) The zero submatrix is square.

21. Let V be an $m \times n$ matrix of the form

$$V = \begin{bmatrix} 1 & t_1 & t_1^2 & \dots & t_1^{n-1} \\ 1 & t_2 & t_2^2 & \dots & t_2^{n-1} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & t_m & t_m^2 & \dots & t_m^{n-1} \end{bmatrix}$$

where $t_1, \dots, t_m$ are numbers. Multiplying an n -vector c by the matrix V is the same as evaluating the polynomial of degree less than n , with coefficients $c_1, \dots, c_n$, at the points $t_1, \dots, t_m$.

- (a) Let $c \in \mathbb{R}^n$ be the coefficient vector. Using the definition of matrix-vector multiplication, explicitly write out the first two entries of the resulting vector Vc to verify it matches $p(t_1)$ and $p(t_2)$.
- (b) Suppose $Vc = 0$ (the zero vector in $\mathbb{R}^m$). What does this imply about the polynomial $p(t)$ evaluated at the points $t_1, \dots, t_m$?
- (c) Show that the columns of V are linearly independent if $m \geq n$ and the numbers $t_1, \dots, t_m$ are distinct, i.e., different from each other.

Note: The matrix V considered in this question is called the **Vandermonde Matrix**. It is a fundamental matrix involved in the applications of polynomial interpolation and curve fitting.